

Macroeconomic and Financial Indicators: Cross-Country Exploratory Analysis

Haoran Li

Introduction

Data source and collection

The data come from the World Bank **World Development Indicators (WDI)** database (World Bank, 2026), which compiles national accounts, survey, and administrative records submitted by member countries to the World Bank, IMF, ILO, and OECD. Indicator codes were checked against the WDI R package (full details in `FindData.Rmd`). Rather than pulling from the live API, this report uses a fixed snapshot (`Dataset.csv`) downloaded from World Bank DataBank on 24 February 2026—covering 1960–2025—so all results can be reproduced offline.

Unit of observation

Each row is a **country–year pair**. Dropping World Bank regional and income-group composites leaves 12,113 observations with at least one recorded value, across 217 economies over 1960–2025. The sample covers UN member states and dependent territories that maintain separate national accounts (e.g., Hong Kong SAR, Virgin Islands U.S.), so “economy” is the more accurate label throughout.

Variables, natural limits, and missing-value coding

All eight indicators are continuous numeric variables; the WDI package codes missing entries as NA rather than sentinel values like `–99`. No cross-variable logical constraints apply—each indicator is defined independently of the others. Table 1 lists each variable alongside its unit and the natural limits any plausible value must satisfy.

Table 1: Variable definitions, units, and natural limits.

Variable	Description	Unit	Natural Limits
Credit/GDP	Domestic credit to the private sector	% of GDP	≥ 0 (no strict upper bound)
GDP Growth	Annual growth rate of real GDP	%	No strict bounds; $ x > 50$ is suspicious
GDP per Capita	Gross domestic product divided by population	Current US\$	> 0
Investment/GDP	Gross capital formation (fixed investment + inventories)	% of GDP	No strict lower bound under SNA 2008; extreme negatives warrant scrutiny
Inflation	Annual change in consumer price index	%	No strict bounds; $x > 500$ suggests hyperinflation
Market Cap/GDP	Market capitalisation of listed domestic companies	% of GDP	≥ 0 (no strict upper bound)
Trade/GDP	Sum of exports and imports of goods and services	% of GDP	≥ 0 (can exceed 100 for entrepôt economies)
Unemployment	Share of the labour force without work (ILO estimate)	%	$0 \leq x \leq 100$

Data Quality Checks

Missing data

Market Cap/GDP leads on missingness at 81.2%—stock-exchange data simply do not exist for countries without formal capital markets. There is also a clear temporal skew: pre-1970 coverage is thin across most variables, a legacy of limited statistical capacity in newly independent states. The gaps are not independent, either. Some 52.8% of observations missing Market Cap/GDP are also missing Credit/GDP. That co-missingness pattern matters practically: financial variables may need to be handled as a group rather than variable-by-variable when imputing or filtering.

Natural-limit violations and outliers

Unemployment and GDP per Capita both respect their natural limits: unemployment stays within $[0, 100]$ and every GDP per Capita entry is positive. That

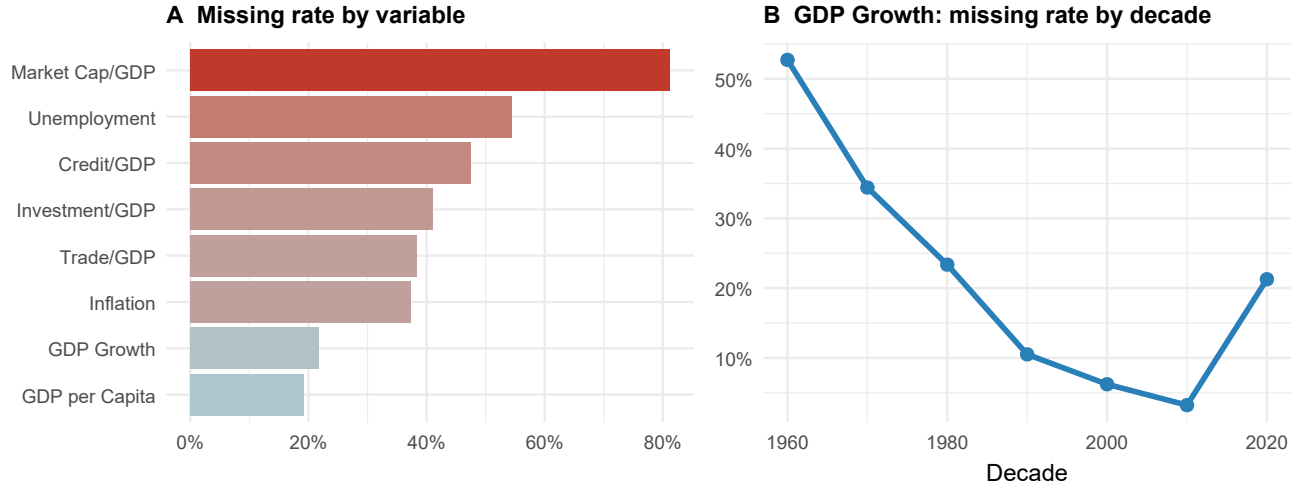


Figure 1: Left: percentage of country–year observations missing for each indicator. Right: share of economies with missing GDP Growth values, grouped by decade—illustrating that data coverage has improved markedly since the 1970s. Both panels are restricted to economy-level observations (aggregates excluded).

said, the lowest recorded value—US\$11.8 for Myanmar in 1966—is almost certainly an exchange-rate artefact tied to early post-independence data, not a real economic figure.

Negative Investment/GDP: 11 observations show negative gross capital formation (lowest: -15.7% in Djibouti, 2021). SNA 2008 permits this when inventory drawdowns or asset disposals outstrip new fixed investment—something that occasionally occurs during post-conflict rebuilding or deep recessions. These values are kept but flagged for extra scrutiny in investment-focused work.

Extreme GDP Growth: 22 observations exceed the $|x| > 50\%$ threshold marked suspicious in Table 1. The single highest—150% in Equatorial Guinea (1997)—fits the familiar pattern of post-conflict rebounds or oil-boom base effects. These are not recording errors, but they carry substantial leverage in any regression and deserve individual inspection.

Extreme Trade/GDP: The highest value is 863% (Virgin Islands (U.S.), 2008). Figures above 400% are quite plausible for small entrepôt economies—Hong Kong SAR, Singapore, Luxembourg—where goods in transit vastly exceed domestic output, making the GDP denominator misleadingly small relative to trade flows.

Extreme Market Cap/GDP: Ratios above 100% arise when large multinationals list domestically but generate revenue abroad. Hong Kong SAR is the clearest example—mainland Chinese firms trade on the Hong Kong exchange yet their output does not enter Hong Kong’s GDP. The ratio is inflated by a denominator mismatch, not bad data.

Hyperinflation: 6 country–year observations record in-

flation above 500%.

Table 2: Country–years with inflation exceeding 500%.

Country	Year	Inflation (%)
Congo, Dem. Rep.	1994	23773.1
Bolivia	1985	11749.6
Peru	1990	7481.7
Ukraine	1993	4734.9
Angola	1996	4145.1
Congo, Dem. Rep.	1992	4129.2

Each entry in Table 2 is arithmetically valid and corresponds to a known currency crisis—the Congo, Dem. Rep. episode in 1994 being the most severe. These observations stay in the dataset, but log-transformation or winsorisation is strongly advisable before fitting any regression.

Data Summaries

Summary statistics

Table 3: Summary statistics for all eight indicators (individual country–years).

Variable	N	Min	Q1	Median	Mean	Q3	Max	NA (%)
Credit/GDP	7523	0.0	12.9	26.3	39.5	51.9	301.0	47.5
GDP Growth	11214	-64.0	1.2	3.8	3.6	6.3	150.0	21.7
GDP per Capita	11569	11.8	613.0	2,195.4	9,658.0	9,404.5	288,001.4	19.2
Inflation	8968	-17.6	2.1	4.8	22.8	10.1	23,773.1	37.4
Investment/GDP	8441	-15.7	18.0	22.7	23.3	27.6	76.8	41.1
Market Cap/GDP	2695	0.0	16.2	34.6	60.3	71.5	1,777.2	81.2
Trade/GDP	8819	0.0	44.6	66.7	78.2	97.0	863.2	38.4
Unemployment	6531	0.1	3.6	6.3	8.1	10.9	38.8	54.4

Distributions

Log-transformation is necessary for two variables (Figure 2). GDP per Capita spans roughly three orders of

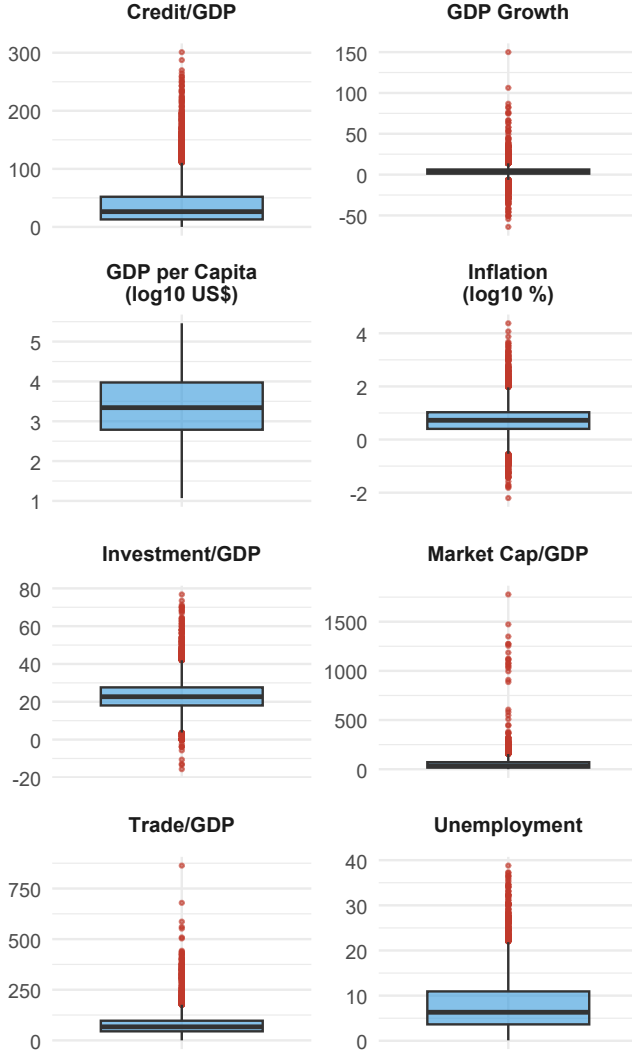


Figure 2: Distribution of each indicator across all country-year observations. GDP per Capita and Inflation are displayed on a log10 scale (positive values only) to handle extreme right skew; negative inflation observations are excluded from those panels only. Boxplot whiskers extend to 1.5 IQR; individual points beyond are drawn in red.

magnitude, and Inflation’s distribution still shows a pronounced right tail even on a log scale—a direct footprint of the hyperinflationary episodes in Table 2. The other six variables sit more symmetrically around their medians, though none are entirely free of outliers.

Time-series trends by income group

Figure 3 surfaces three patterns worth flagging for subsequent modelling. Financial deepening (Credit/GDP) has climbed steadily in high-income countries since 2000 while barely moving in low-income ones—the gap keeps widening. Trade/GDP tells a more cyclical story: it rose across all groups through the 1990s–2000s, flattened after 2008, and has only partially recovered, matching the global trade slowdown documented by Constantinescu, Mattoo & Ruta (2020). Most striking is the GDP growth panel: both crisis windows generate sharp, synchronised contractions across all income groups, and the 2020 COVID-19 shock is by some margin the deepest in the sample.

Bivariate relationships

Figure 4 shows Credit/GDP and GDP per Capita as the tightest pair ($\rho = 0.74$), with Market Cap/GDP and Credit/GDP close behind ($\rho = 0.68$)—the familiar clustering of financial depth and per-capita wealth identified by King & Levine (1993) and Levine (2005). Inflation and GDP Growth show essentially no rank correlation ($\rho = 0.03$), suggesting that the classical negative inflation–growth link operates primarily through extreme episodes rather than across the full distribution. What stands out by contrast is Trade/GDP: it is nearly uncorrelated with the financial cluster, which makes sense given that trade openness is shaped mainly by geography and commercial policy rather than domestic financial conditions.

Conclusions

The dataset covers 217 economies from 1960 to 2025, and the eight indicators examined here are far from clean. The most consequential quality issue is the non-random absence of financial variables: Market Cap/GDP is missing in 81.2% of country-years, and those gaps are concentrated in lower-income economies. Any model that includes these regressors implicitly conditions on the existence of a functioning capital market—a selection that should be made explicit rather than left buried in the sample.

Two distributional features will affect most modelling choices. GDP per Capita and Inflation are both heavily right-skewed, spanning several orders of magnitude; log-transformation is not optional for either. For the remaining six variables the distributions are broadly symmetric, but outliers are common—particularly the hyper-

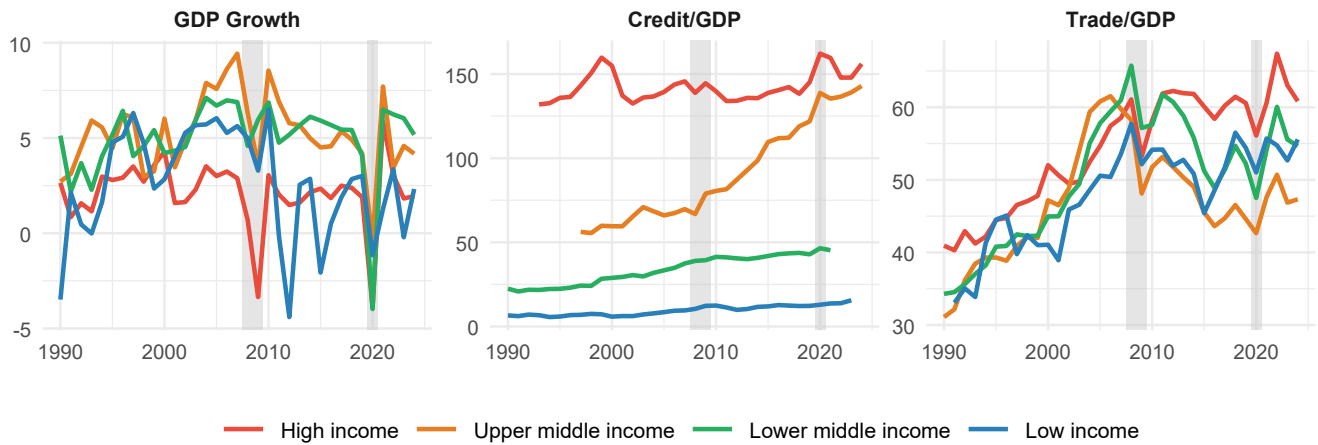


Figure 3: Trends (1990–2023) for three indicators across World Bank income groups. Grey bands mark the Global Financial Crisis (2008–2009) and the COVID-19 shock (2020). Income-group series are World Bank composite aggregates and do not require individual-country imputation.



Figure 4: Spearman rank correlation matrix computed on pairwise-complete country-year observations. Spearman is used in preference to Pearson because GDP per Capita and Inflation are strongly right-skewed. Variables are reordered by hierarchical clustering. Blank cells indicate correlations not reaching significance after Bonferroni correction ($\alpha = 0.05/28$ pairs). Colour intensity reflects magnitude.

inflationary episodes in Table 2, which deserve individual case review before any regression is run.

Structurally, financial depth (Credit/GDP, Market Cap/GDP) tracks closely with per-capita income across the sample, and the income-group time series shows this gap widening since the 1990s—a latent financial development dimension that PCA or factor analysis could usefully extract. Researchers building causal models from this dataset, in the tradition of cross-country growth empirics (Dollar & Kraay, 2002), should plan for two-way fixed effects, explicit treatment of financial-variable missings, and careful handling of hyperinflationary observations. Extending the dataset with institutional indicators (Acemoglu, Johnson & Robinson, 2001) would also help separate financial development from broader governance effects.

References

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